Coding & Solution Report

Development Phase Report - Rising Waters

1. Directory Layout & Architecture

The source code has been structured at the root of the project to run locally or as a Vercel serverless function:

- `app.py`: Local web server.
- `api/index.py`: Entry point for Vercel.
- `train.py`: Data preprocessing, outlier treatments, and algorithm training.
- `notebook.ipynb`: Interactive notebook documenting the EDA visualizations and confusion matrices.
- `templates/` & `static/`: Layouts and style sheets.

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2. Feature Scaling Parameters

Standardization is computed manually during inference to bypass dependencies. We use parameters fitted on the 75% training split:

- Mean (Average values):

`[36.52325581, 2908.91046512, 28.64447674, 381.60174419, 1999.84011628]`

- Scale (Standard deviation):

`[4.33677284, 436.58575519, 21.71927407, 147.98425606, 380.13250433]`

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3. Decision Tree Compiled Model

The Decision Tree achieved 96.55% accuracy on testing splits. The model splits on June-September monsoon precipitation:

```
# Threshold calculated: 2425.64 mm of Jun-Sep rainfall
if jun_sep_rain > 2425.64:
    # High Flood Risk
    prediction = 1
else:
    # Low Flood Risk (Safe)
    prediction = 0
```

To output smooth, non-static probability scores, a sigmoid logistic scaling function is applied to the threshold difference:

```
diff = jun_sep_rain - 2425.64
prob_raw = 1.0 / (1.0 + math.exp(-diff / 100.0))
```

This enables the UI to animate progress bars indicating realistic model confidence (e.g. 96.5% confidence) based on how far the input values are from the danger threshold.